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Payroll Management System

Data Structure and Programming II | Group I2-GIC-2
Academic Year 2025-2026

Team 3

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Our Team Members



**Sean
Manutnithya**



**Taing
Chan Dara**



**Tang
Sovanoudom**



**Thon
Bounthet**



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TEAM MEMBER CONTRIBUTIONS

Members' Name

Role

SEAN MANUTNITHYA

Leader / Dividing Task / Error checks and fixes / Record, Search and Sort / Time and attendance / Edit Video

TAING CHAN DARA

Program Menu / Tax information / Salary and Tax deduction / Calculate Salary after Deduction / Error checks and fixes

TANG SOVANOUDOM

Project Report / Payment and Pay Schedule / Search and Sort Algorithm / Error fixes

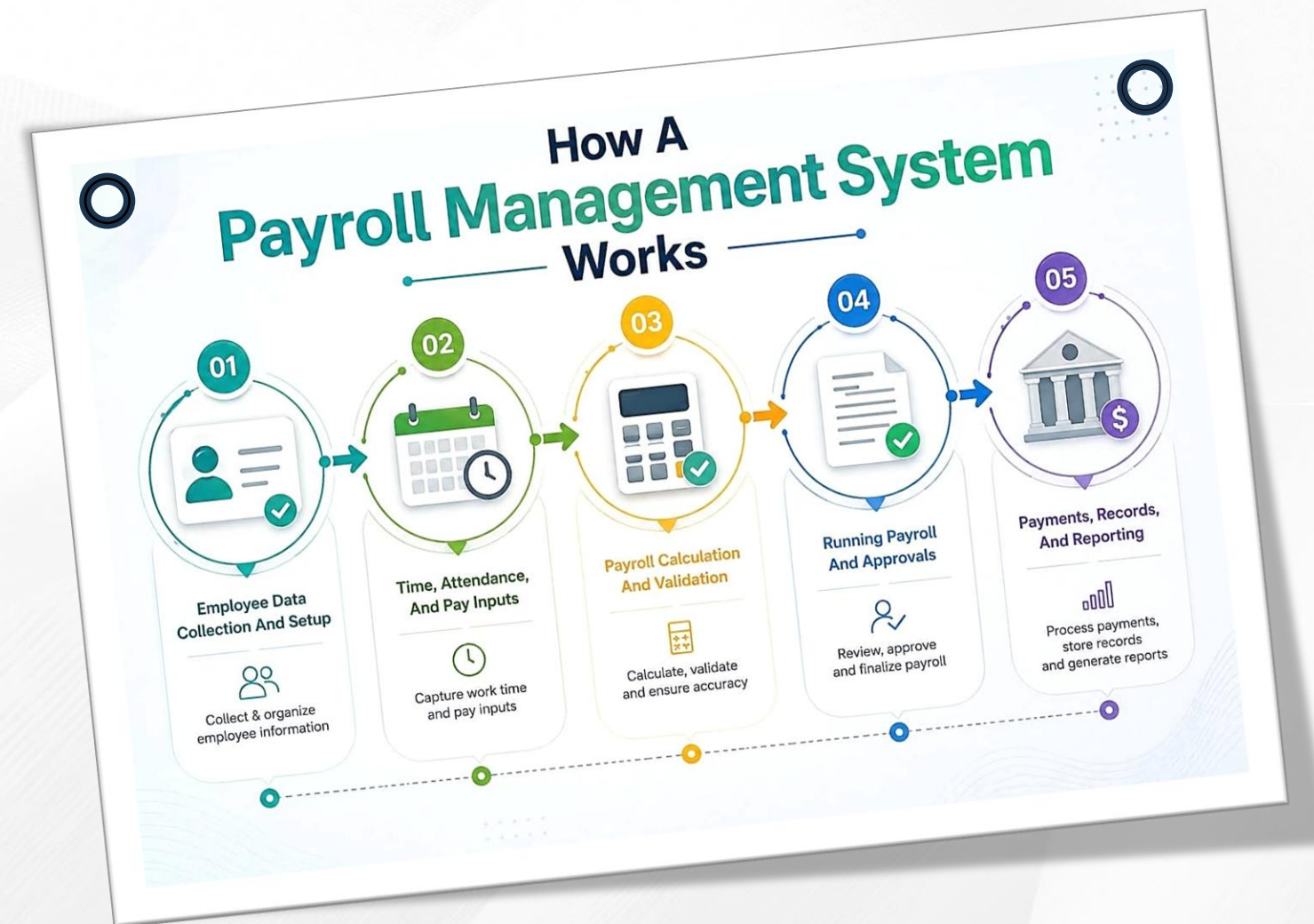
THON BOUNTHET

Project Slide / Calculate Salary after Deduction / Payroll Submission to government / Error fixes



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04. Data Structure & Logic

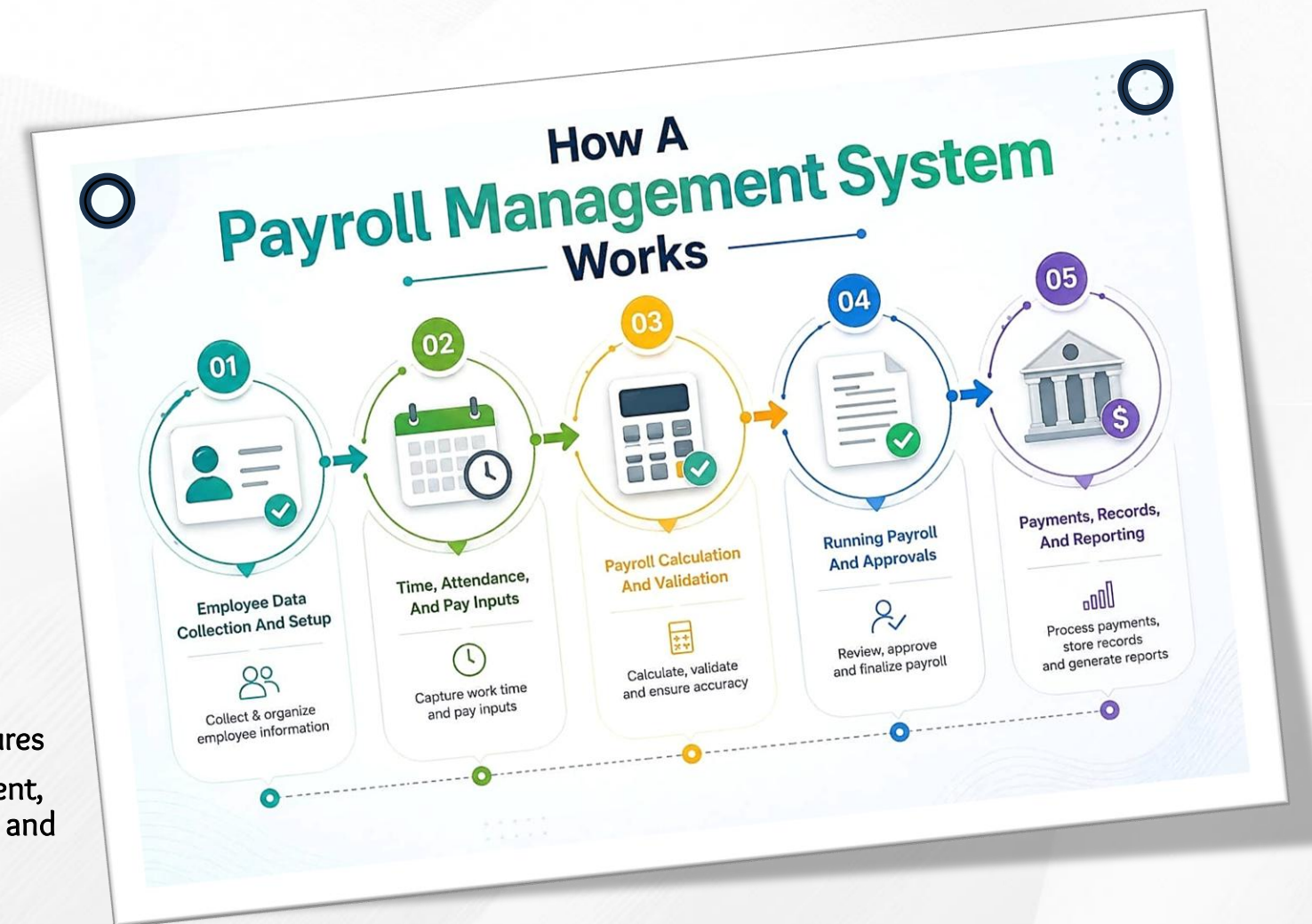
- Doubly Linked Lists, CSV File handling, and algorithms.

02. Core System Features

- Employee management, attendance tracking, and search functions.

03. Technical Implementation

- Programming Language environment and key C++ Modules.



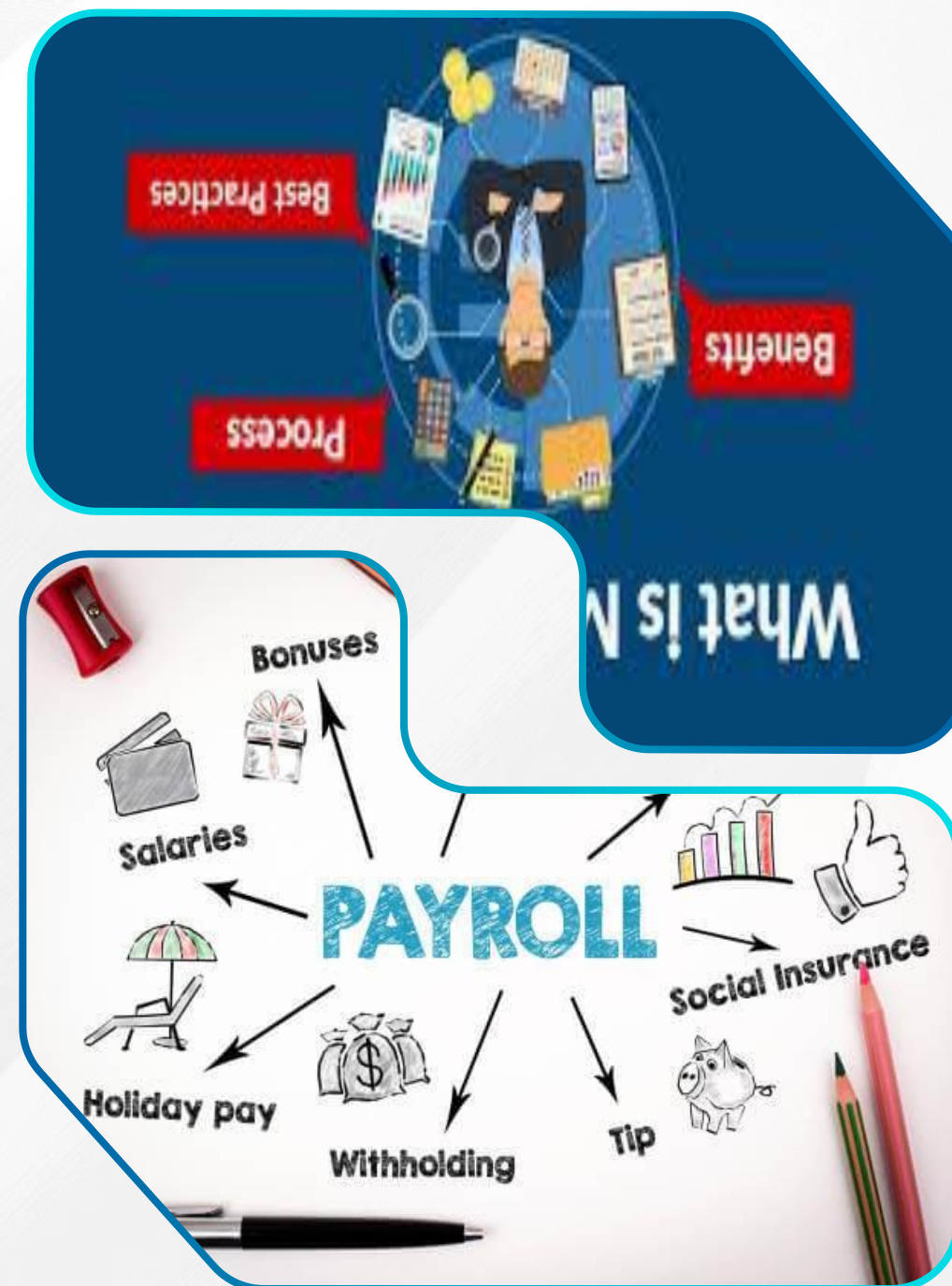


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Introduction & Purpose

Payroll management Payroll management is a critical organizational process. Our C++ based system automates complex tasks to ensure accuracy in compensation.

- 🖥️ Automated working hours tracking.
- 📊 Precise tax and overtime deduction.
- 💾 Lightweight CSV file data persistence.





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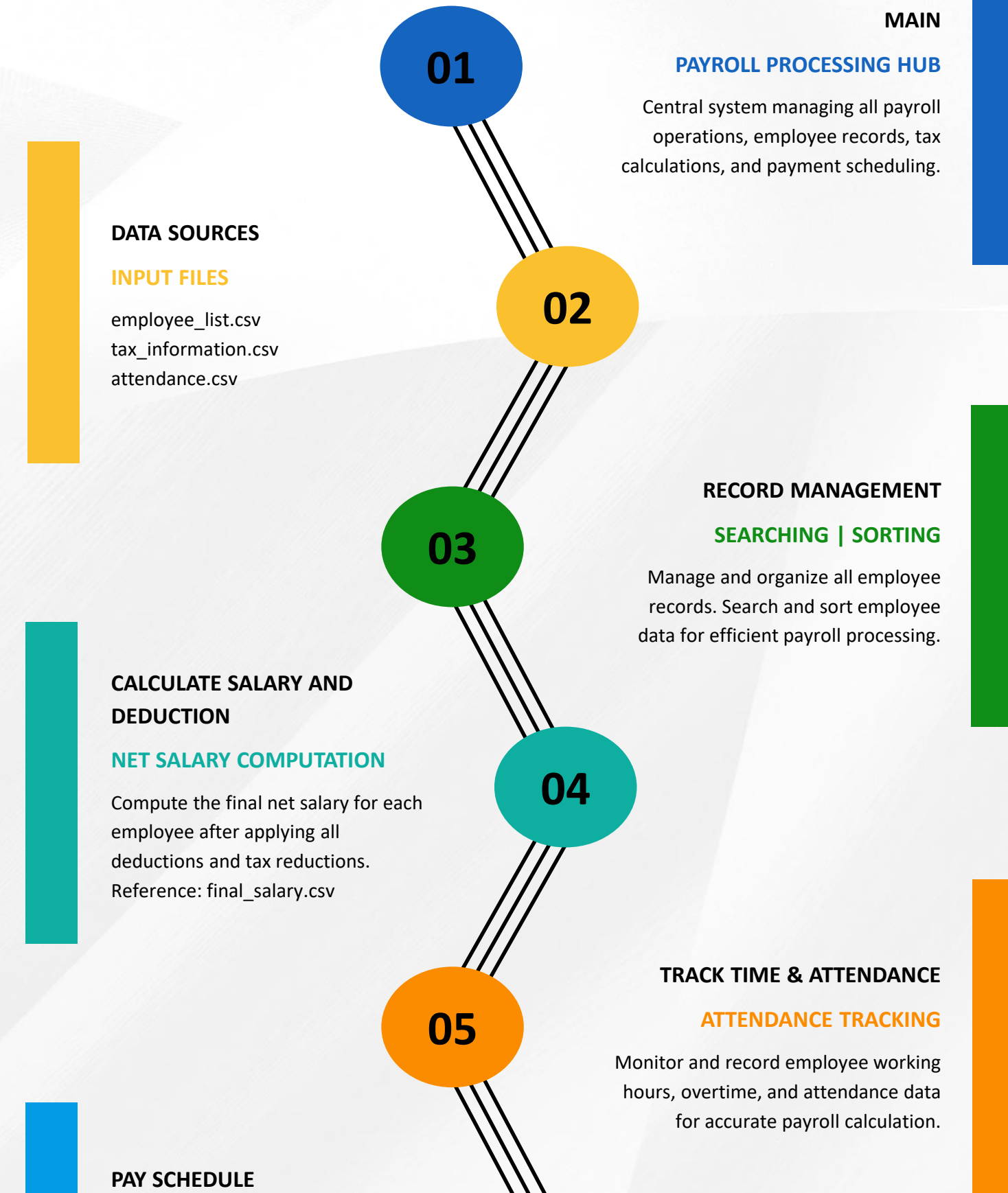
Core System Features

Employee Management

The foundation of the system is the Employee Record module, which maintains essential data:

- Unique Employee ID and Name validation.
- Role-based attributes (Position, Dept).
- Wage parameters (Monthly Salary, Hourly Rate).
- Contact Info (Phone Number, Gender).

PAYROLL MANAGEMENT SYSTEM





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Core System Features

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- Wage parameters (Monthly Salary, Hourly Rate).
- Contact Info (Phone Number, Gender).

PAY SCHEDULE

PAYMENT PLANNING

Organize and schedule payment dates for all employees. Manage pay periods and ensure timely disbursement of salaries.

PAY EMPLOYEES

SALARY DISBURSEMENT

Execute salary payments to all employees via direct deposit or check. Confirm successful payment transactions.

SUBMIT PAYROLL TAX

TAX FILING & PAYMENT

Submit payroll tax payments and filings to relevant authorities.
Reference: tax_report.csv | tax_filling_history.csv

TRACK TIME & ATTENDANCE

ATTENDANCE TRACKING

Monitor and record employee working hours, overtime, and attendance data for accurate payroll calculation.

TAX INFORMATION

TAX PROCESSING

Retrieve and process tax information for each employee including federal, state, and local tax rates.

CALCULATE REDUCTION

DEDUCTIONS & BENEFITS

Calculate all applicable deductions including taxes, benefits, insurance, and other withholdings from gross salary.



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Tracking & Searching

Attendance Tracking

Records working hours, presence, and overtime. Automatically updates employee stats and prepares values for the final payroll calculation.

Linear Search Function

Quickly locate records using Employee ID. The system traverses the linked list to display relevant payroll and personal details instantly.





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Technical Implementation

The system is built as a console-based C++ application, prioritizing performance and modularity.

- Language: C++ (Double Linked List and File IO).
- Modules: Separate headers for Records, Attendance, and Tax Calculation.
- Persistence: CSV format for lightweight data storage and portability.

```
1 #include <iostream>
2 #include <vector>
3
4 // Define the Student class
5 class Student {
6 public:
7     Student() {
8         // Initialize numbers in the constructor
9         numbers = {10, 21, 35, 48, 56};
10    }
11
12    // Method to get the list of numbers
13    std::vector<int> getNumbers() {
14        return numbers;
15    }
16
17 private:
18     std::vector<int> numbers;
19 };
20
21 int main() {
22     std::cout << "---- Welcome to C++ Program ----" << std::endl;
23     std::cout << "Hello, C++ World!" << std::endl;
24
25     Student alex;
26     std::vector<int> numbers = alex.getNumbers();
27
28     std::cout << "Numbers in vector: ";
29     for (int i = 0; i < numbers.size(); ++i) {
30         std::cout << numbers[i] << " ";
31     }
32     std::cout << std::endl;
33
34     std::cout << "Program executed successfully." << std::endl;
35     return 0;
36 }
```

Console/Terminal x Bash

```
[Build] Successful. (1.2s)
alex@programmer:~/MyProject/build$ ./MyProject
---- Welcome to C++ Program ----
Hello, C++ World!
Numbers in vector: 10 21 35 48 56
Program executed successfully.
```




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Data Structure & Logic

Doubly Linked List Logic

We chose the Doubly Linked List for its flexibility in managing employee nodes:

- 01** Dynamic Memory:
Scales without fixed-size arrays.
- 02** Bidirectional Traversal:
Easy access to Next and Previous nodes.
- 03** Efficiency:
Constant time complexity for insertion and deletion of staff records.

```
payroll_system-main > payroll_system-main > payroll_s
1 1 ▾ #ifndef LINKED_LIST_H
2   #define LINKED_LIST_H
3
4 2 ▾ #include <iostream>
5   #include <fstream>
6   #include <sstream>
7   #include <string>
8   #include <iomanip>
9
10  #using namespace std;
11
12 3 ▾ struct Employee_info {
13     string ID;
14     string Name;
15     float Baseslary;
16     float Tax_salary;
17     float Tax_Bracket;
18     float Tax_Amount;
19     float Socialsociety;
20     float Overtime_Hours;
21     float Overtime_Amount;
22     float Absence_Count;
23     float Absence_Deduction;
24     float Total_deduction;
25     float Main_salary;
26
27     string Salaryopendate;
28     string payrollsubmitdate;
29 };
30
31 4 ▾ struct Box {
32     Employee_info data;
33     Box* next;
34 };
35
36 #endif
```

```
#ifndef STRUCTURE_H
#define STRUCTURE_H
#include <iostream>
using namespace std;
struct Attendance_record{

    string ID;
    string name;
    string gender;
    string department;
    string position;
    string salary;
    string hourly_paid;
    string phone;

    string overtime;
    string attendance_count;
    string work_hour;
    string overtime_paid;
    string total_cost;

    Attendance_record * next;
    Attendance_record * prev;

};

struct Employee{
    int size;

    Attendance_record * head;
    Attendance_record * tail;
};

struct Payday {
    int workingDay;
    string name;
    double pay;
};

struct Node {
    int workingDay;
    string name;
    double payment;
    Payday data;
    Node* next;
};

#endif
```




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File IO

Reading Files

Employee and attendance data are loaded from CSV files into linked lists during program initialization. The system automatically reads existing records at startup, restoring all employee profiles, attendance logs, and payroll data into memory for immediate use.

Writing Files

Updated records are saved back to their respective CSV files after any modifications are made. This ensures full data persistence across sessions — all changes to employee info, attendance, and payroll calculations are retained even after the program closes.



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System Result

Processing Algorithms

Linear Search

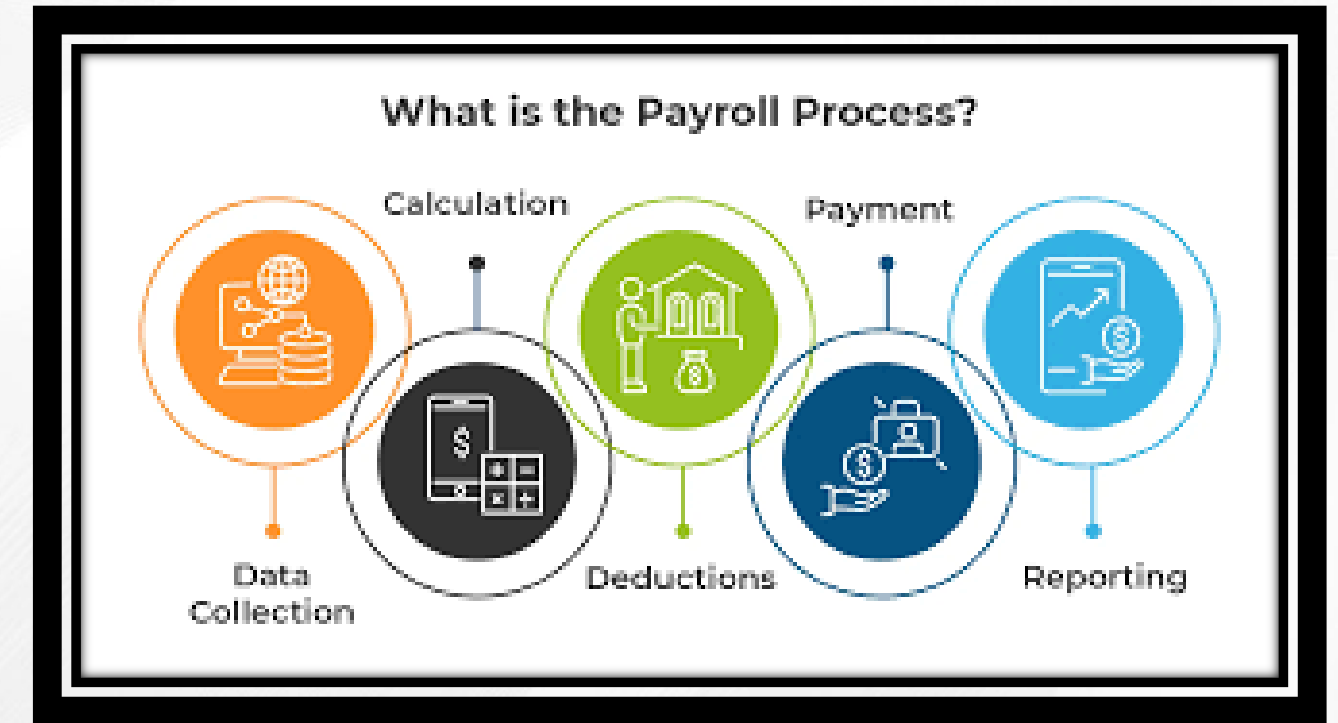
Traverses the list from head to tail to match target IDs for record retrieval.

Bubble Sort

Organizes attendance data in Ascending or Descending order for performance reports.

File Handling

Reads from employee_list.csv and writes back updated payroll information.



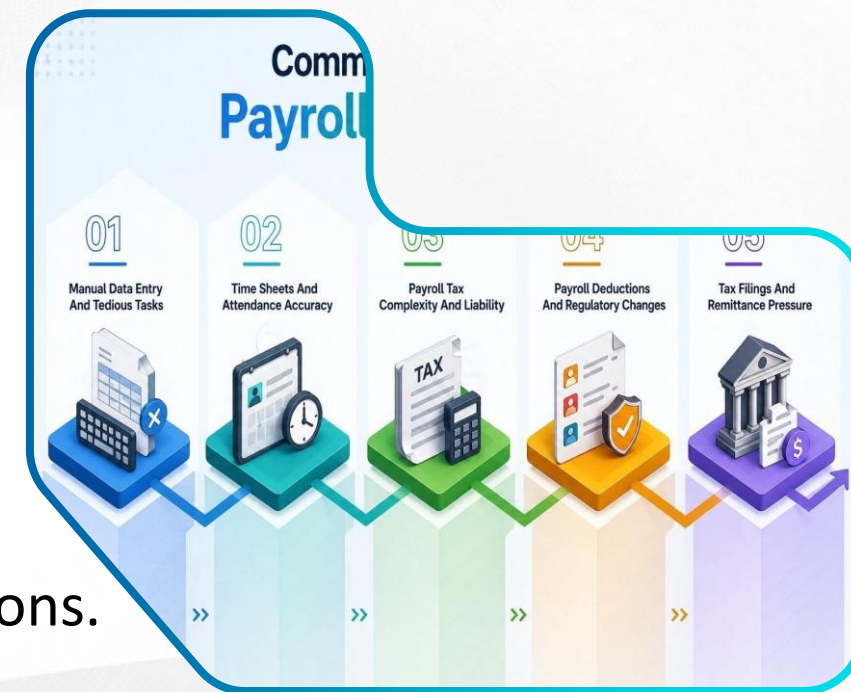


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SALARY & TAX CALCULATION

The core logic integrates earnings with government tax regulations and attendance deductions.

$$\text{Salary}_{\text{Final}} = \text{Salary}_{\text{Base}} + \text{Pay}_{\text{OT}} - (\text{Tax}_{\text{Amount}} + \text{Absence}_{\text{Dedu}} + \text{Social}_{\text{society}})$$



INPUT DATA

BASIC SALARY
WORKING HOURS
OVERTIME HOURS
TAX RATE

ATTENDANCE

ATTENDANCE COUNT
OVERTIME = HOURS × HOURLY RATE
ATTENDANCE DEDUCTION APPLIED

CALCULATION

TAX AMOUNT = TAX_SALARY ×
(TAX_RATE_BRACKET/100)
AUTOMATED COMPUTATION

OUTPUT

FINAL SALARY = BASIC + OVERTIME -
TAX_AMOUNT - ABSENCE - NSSF



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System Result & Validation

Test Case Success

Extensive testing ensured that the system handles edge cases effectively:

- ☐ Accurate processing of large attendance CSVs.
- ☐ Proper tax application across different salary tiers.
- ☐ Validation of ID search and sorting results.
- ☐ Successful "Submit Payroll" file persistence.



Verified CSV Output
final_Salary.csv



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Learning Outcomes

“Combining Program Logic with administrative efficiency to create meaningful software solution”

Conclusion & Learning

Summary of Achievements

The project successfully integrated data structures with real-world business requirements.

- 01 Practical mastery of Linked Lists.
- 02 Robust understanding of File I/O in C++.
- 03 Enhanced problem-solving and logic skills.
- 04 Team coordination and modular design.



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Demo code

```
PS C:\Users\ASUS ZenBook\Downloads\Project\Project\payroll_system-main\payroll_system-main> cd "c:\Users\ASUS ZenBook\Downloads\Project\Project\payroll_system-main\payroll_system-main\" ; if ($?) { g++ menu.cpp -o menu } ; if ($?) { .\menu }
File is not open!
=====Payroll Management=====
1 --> Record Attendance for the month
2 --> Search
3 --> Tax information
4 --> Calculate Deduction & Final Salary
5 --> Sort
6 --> Pay Schedule
7 --> Submit Payroll
0 --> Exit Program
=====
Enter your choice: 
```




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**Thank
You**
For Your Attention

